

سوال نمبر ۱۰

① An EM wave travels in free space in z direction with $\underline{E}_z = 300 \text{ V/m}$

② The amplitude of B in free space

$$\therefore c = \frac{E}{B} \rightarrow B = \frac{E}{c} = \frac{300}{3 \times 10^8} = 1 \times 10^{-6} \text{ T (y)}$$

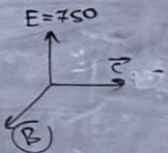
③ The amplitude of B in medium the speed is 80% of the speed of light

$$v = \frac{80}{100} \times 3 \times 10^8 = 2.4 \times 10^8$$

$$B = \frac{E}{v} = \frac{300}{2.4 \times 10^8} = 1.25 \times 10^{-6} \text{ T (y)}$$

سے العمل کرتا ہے

② EM wave travels in free space with $E_y = 750 \text{ V/m}$. The wavelength and B in z with 40 MHz



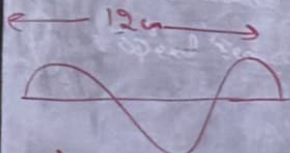
$$B = \frac{E}{c} = \frac{750}{3 \times 10^8} = 2.5 \times 10^{-6} \text{ T (z)}$$

$$\lambda = \frac{c}{f} = \frac{300 \text{ M}}{40 \text{ M}} = 7.5 \text{ m}$$

$$\frac{V}{d} = E = \frac{F}{q}$$

$$\frac{F}{q} = \frac{V}{d} \Rightarrow F \cdot d = q \cdot V$$

$$W = K E$$



$$\lambda = \frac{12}{1.5} = 8 \text{ cm}$$

سے الیٹریکٹیویٹی

③

$$\sin(kx - \omega t)$$

Free space with: $E_y = 600 \sin(10^7 x - \omega t)$

$$c = \frac{\omega}{k}$$

$$E_{\max} = B_{\max} \cdot c$$

$$B_{\max} = \frac{E_{\max}}{c}$$

$$E_z = 300 \sin(kx - \omega t)$$

$$B_y = \frac{10^{-6}}{10} \sin(kx - \omega t)$$

Find

① Magnetic field density $\rightarrow B$

$$B = \frac{E}{c} = \frac{600}{3 \times 10^8} = 2 \times 10^{-6} \text{ at } z \rightarrow B = 2 \times 10^{-6} \sin(10^7 x - \omega t)$$

② Magnetic field Intensity $\rightarrow H$

$$H = \frac{B}{\mu} \rightarrow \frac{2 \times 10^{-6}}{4\pi \times 10^{-7}} = 1.6 \text{ at } z \rightarrow H = 1.6 \sin(10^7 x - \omega t)$$

سید الفیاضی

③

$$\sin(kx - \omega t)$$

free space with: $E_y = 600 \sin(10^7 x - \omega t)$

$$c = \frac{E}{B}$$

$$E_{max} = B_{max} \cdot c$$

$$B_{max} = \frac{E_{max}}{c}$$

$$B_z = 0.3 \sin(kx - \omega t)$$

$$E_y = 0.9 \times 10^8 \sin(kx - \omega t)$$

Find

① Magnetic field density $\rightarrow B$

$$B = \frac{E}{c} = \frac{600}{3 \times 10^8} = 2 \times 10^{-6} \text{ at } z \rightarrow B = 2 \times 10^{-6} \sin(10^7 x - \omega t)$$

② Magnetic field Intensity $\rightarrow H$

$$H = \frac{B}{\mu} \rightarrow \frac{2 \times 10^{-6}}{4\pi \times 10^{-7}} = 1.6 \text{ at } z \rightarrow H = 1.6 \sin(10^7 x - \omega t)$$

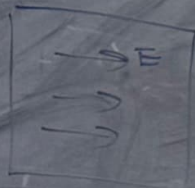
مسئله ۱۰

$$\vec{E} = 2\hat{i} - \hat{j} + 3\hat{k}$$

a) $\vec{B} = -2\hat{i} + \hat{j} - 3\hat{k}$

b) $\vec{B} = 3\hat{i} + 2\hat{k}$

c) $\vec{B} = -\hat{i} - 2\hat{j}$



if $\vec{E} \times \vec{B} = 0$
then $\vec{E} \parallel \vec{B}$